Exp No:5 Data Collection and Initial Exploration

AIM:

To collect a sales dataset in CSV format and explore the data using Python. To identify missing values and visualize sales data using graphs.

PROGRAM:

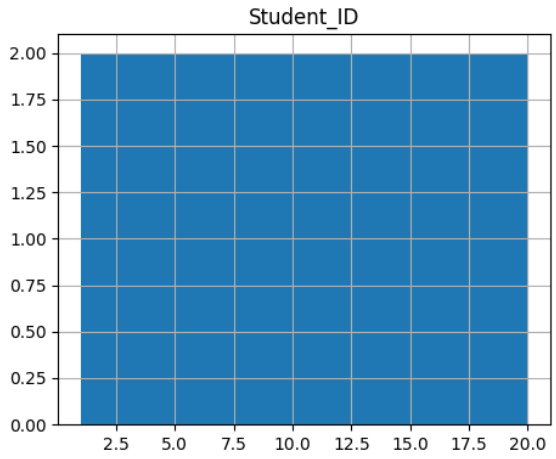
```
from google.colab import files
uploaded = files.upload()
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
df = pd.read_csv("student_data.csv")
print(df.head())
print(df.info())
print(df.describe())
df.hist(bins=10, figsize=(12, 15))
plt.show()
sns.pairplot(df)
plt.show()
```

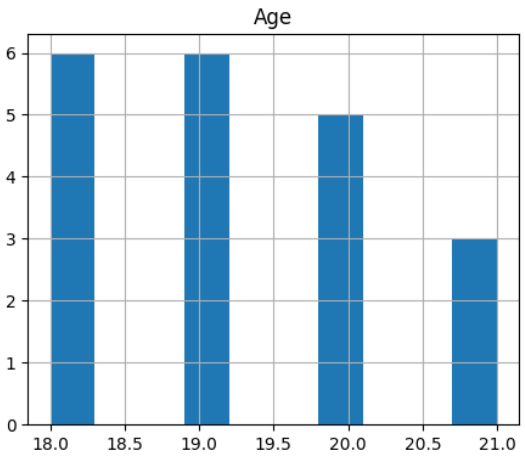
OUTPUT:

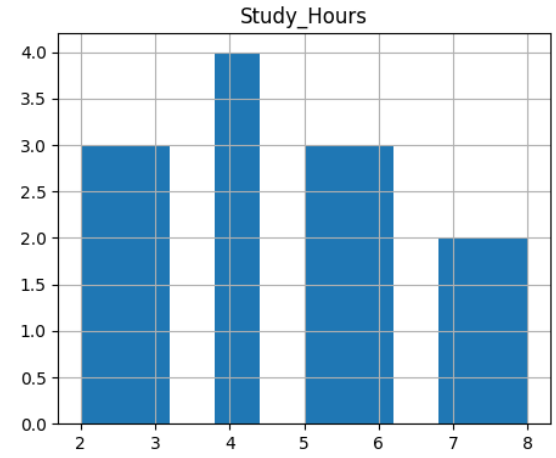
```
  Student_ID  Age  Study_Hours  Attendance  Marks  Assignments_Completed
0          1   18            2          75     55                     6
1          2   19            4          82     68                     8
2          3   18            3          78     60                     7
3          4   20            6          90     85                    10
4          5   19            5          85     76                     9
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 20 entries, 0 to 19
Data columns (total 6 columns):
 #   Column               Non-Null Count  Dtype
---  ------               --------------  -----
 0   Student_ID           20 non-null     int64
 1   Age                  20 non-null     int64
 2   Study_Hours          20 non-null     int64
 3   Attendance           20 non-null     int64
 4   Marks                20 non-null     int64
 5   Assignments_Completed  20 non-null     int64
dtypes: int64(6)
memory usage: 1.1 KB
None
       Student_ID        Age  Study_Hours  Attendance       Marks  \
count    20.00000  20.000000     20.00000      20.000  20.000000
mean     10.50000  19.250000      4.70000      84.750  74.300000
std       5.91608   1.069924      1.92217       8.309  14.733063
min       1.00000  18.000000      2.00000      70.000  50.000000
25%       5.75000  18.000000      3.00000      78.750  61.500000
```

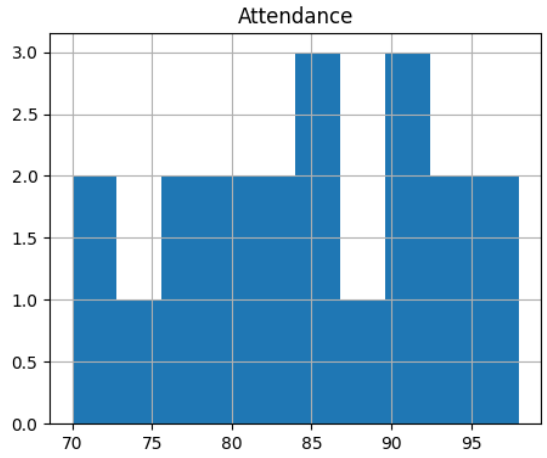
| | | | | | |
|---|---|---|---|---|---|
| 50% | 10.50000 | 19.000000 | 4.50000 | 84.500 | 74.500000 |
| 75% | 15.25000 | 20.000000 | 6.00000 | 91.250 | 87.250000 |
| max | 20.00000 | 21.000000 | 8.00000 | 98.000 | 96.000000 |

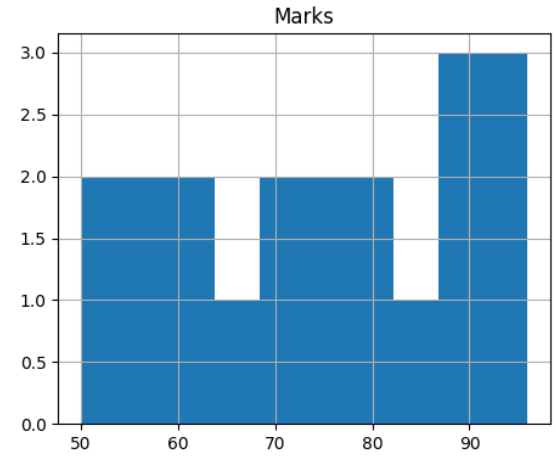
| | Assignments_Completed |
|---|---|
| count | 20.000000 |
| mean | 8.250000 |
| std | 1.743409 |
| min | 5.000000 |
| 25% | 7.000000 |
| 50% | 8.500000 |
| 75% | 10.000000 |
| max | 10.000000 |

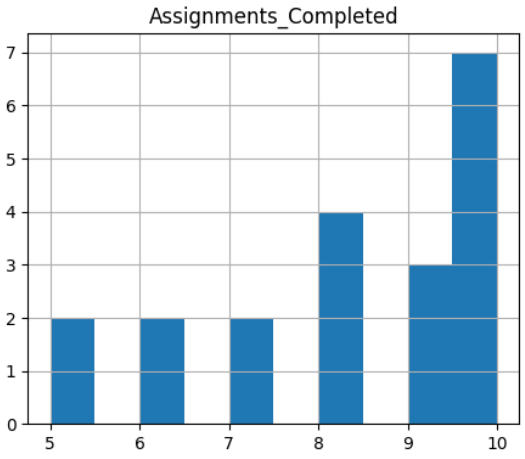

Student_ID
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

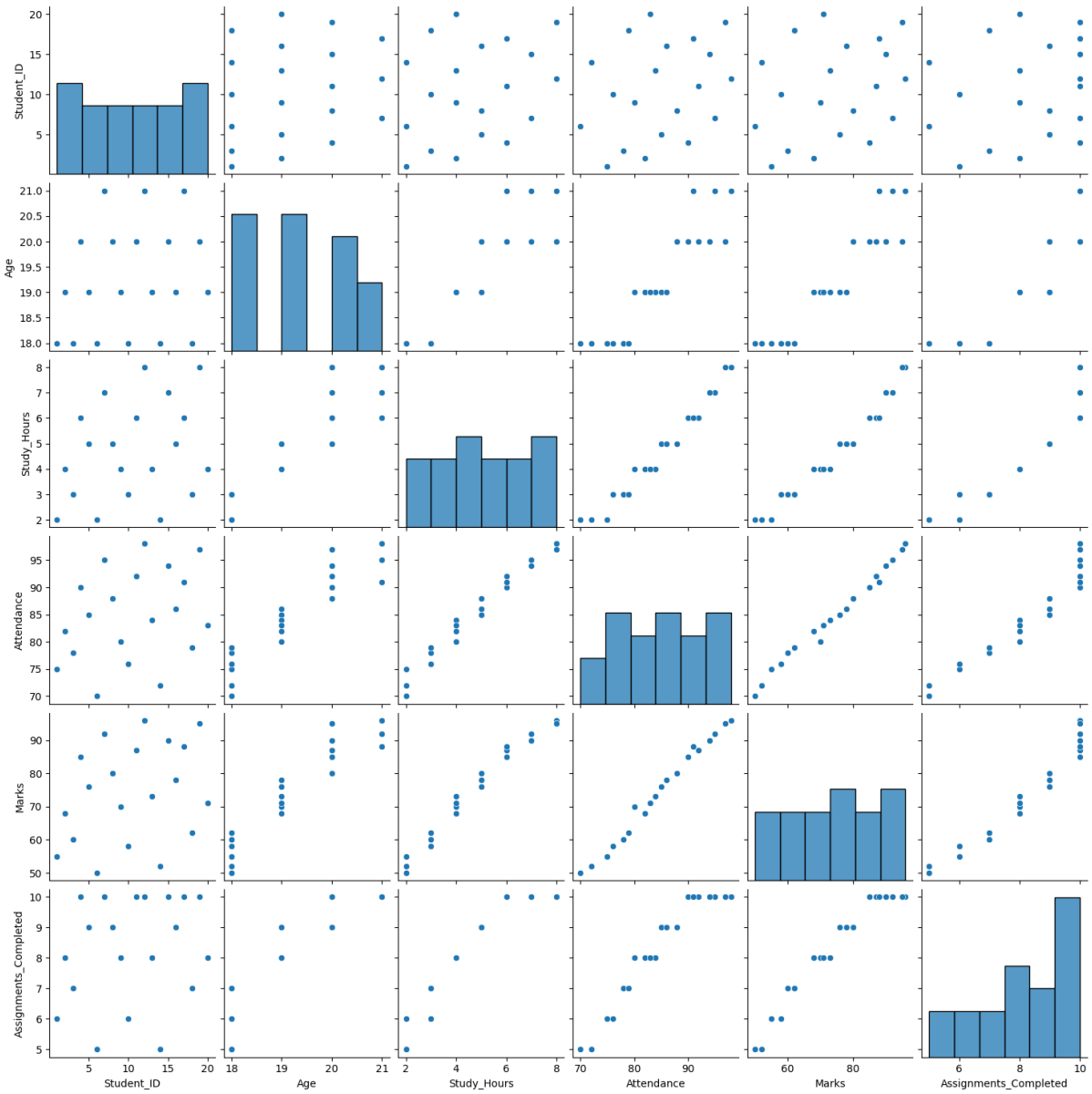

Age
6
5
4
3
2
1
0
18.0
18.5
19.0
19.5
20.0
20.5
21.0


Study_Hours
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
2
3
4
5
6
7
8


Attendance
3.0
2.5
2.0
1.5
1.0
0.5
0.0
70
75
80
85
90
95


Marks
3.0
2.5
2.0
1.5
1.0
0.5
0.0
50
60
70
80
90


Assignments_Completed
7
6
5
4
3
2
1
0
5
6
7
8
9
10




RESULT:

The student dataset was successfully loaded and explored using Pandas. Summary statistics and visualizations were generated to understand the distribution and relationships among the student data.